

Maedeh Dehghan

 GitHub |  LinkedIn |  maedeh3@ualberta.ca |  maedehdehghanam.github.io

Skills

Programming Languages: Python, Java, Golang, MATLAB, C/C++, HTML, CSS
Tools & Libraries: Git, Linux, PyTorch, Pandas, Numpy, scikit-learn, LLVM, Docker, Kubernetes
Hardware Description Languages: VHDL, SystemVerilog
Assembly Languages 8086, ARM, RISC-V

Work Experience

Xanadu May. 2026 - Aug. 2026

Resident | Full-time

- Enhancing the JIT compilation toolchain (Catalyst) by leveraging MLIR dialects to streamline the transformation of quantum kernels, increasing the scalability of large-scale circuit simulations.

Amii (Alberta Machine Intelligence Institute) Oct. 2025 - Nov. 2025

Technical Advisement (WILO) | Part-time

- Contributed to brainstorming ML approaches for solving the client's business problem.
- Conducted academic and industrial literature reviews to determine ML approaches for solving the client's business problem.

Institute For Research In Fundamental Sciences (HPC Center) Oct 2023 - Jun 2024

Platform Engineer | Part-time

- Built a Kubernetes platform for users to launch Google-Colab-like notebook environments.
- Implemented dynamic resource allocation across CPUs, GPUs, and memory in the HPC cluster.
- Automated deployment and scaling workflows to improve usability and resource efficiency.

Farzanegan Highschool Sep. 2020 - Feb. 2023

Physics and Calculus Instructor | Part-time

- Taught physics and calculus to students preparing for national Olympiads.
- Designed advanced problem-solving lessons and mentored high-achieving students.

Education

University of Alberta Sep. 2024 - Dec. 2026 [Expected]

Master of Science in Computer Science **GPA: 3.8/4**
Developed an ML-based selection framework to optimize DNN inference by dynamically identifying the most efficient implementation primitives for Convolutional Neural Network layers.

Shahid Beheshti University Sep. 2020 - Jun. 2024

Bachelor of Science in Computer Engineering **GPA: 3.9/4**
Relevant Coursework: Compiler Design, Operating Systems, Microprocessor & Assembly Languages

Project Work

LLVM IR Optimization using Large Language Models

LLVM, Clang, Hugging Face

- Implemented tools in Python to parse, transform, and apply loop optimizations to LLVM Intermediate Representations (LLVM IR).
- Explored how Large Language Models can analyze and optimize LLVM IR.

Multi-Cycle RISC-V Processor

SystemVerilog, ModelSim

- The project involves the design and implementation of a multi-cycle RISC-V processor.
- The basic state machine used in the project is a Moore machine that takes [6:0]op as input and produces sets of outputs.

Laptop Price Prediction

Python, Selenium, Pandas, Numpy, scikit-learn

- This project is a machine learning project that predicts laptop prices based on Torob.com.
- Applied **K-Fold Cross Validation technique** to enhance model performance.

Teaching Assistant Experience

Computer Organization and Architecture I

Fall 2025

University of Alberta

Operating Systems (x3)

Spring 2023 - Spring 2024

Shahid Beheshti University

Spring 2022 - Spring 2024

Advanced Programming (x4)

Shahid Beheshti University

Fall 2022 - Spring 2023

Electrical Circuits (x2)

Shahid Beheshti University

Fall 2023

Algorithm Design

Shahid Beheshti University

Spring 2023

Artificial Intelligence

Shahid Beheshti University

Honors

- **Ranked 3rd** among 92 B.Sc. students in the Computer Engineering department, Shahid Beheshti University, 2023.
- **Ranked 590th** in National Entrance Exam for B.Sc Studies among 160,000 students, 2020.
- **Semifinalist** National Physics Olympiads, 2018 & 2019.